

Post-Retirement Part I: Annuities

The Haves and Have-Nots in the Annuities Space

Notes from inside the pricing model. What we have, what we are missing and what it would take to close the gap

One of the investments I have been thinking about lately is an annuity. If you ever take a peek at how an annuity is priced, you start to appreciate how the power of compounding shows up. I do not have the funds to buy one yet, so I figured I would spend the wait doing some research so that when the time comes, I will know exactly what I want. The further I get, the clearer one thing becomes: the annuity I want does not exist in my market. So rather than travel elsewhere to buy it, I thought, why not try to bring it here? Or at least inspire it here. And if that is not possible, then let us at least be honest about what we have, what we are missing and what it would take to close the gap. Here are my two cents.

Having looked at an annuity model, I think we have what the model demands and what the assumptions permit, the actuarial machinery itself is sound. The model asks the right questions and our practice answers them. The problem sits one layer down, in what the answers are actually calibrated to and what they assume about a future we have not yet measured. But pay attention to the distance between what our model assumes and what global practice does, and the mismatch is real. The East African annuity market is functional. Products exist, capable actuaries price them and the regulators presumably do the best they can. What is less visible, from the outside and sometimes from the inside, is the build-up of structural misalignments between how we price these products and what the underlying risks actually are. The simplest place to start is with the question every annuity pricing model has to answer first. Who do we think we are pricing for?

Kenya has commissioned two domestic mortality studies in the last twenty years. Association of Kenya Insurers (AKI) led both, with Zamara as the consultancy. The first produced the KE 2001–2003 tables in 2007; the second covered the 2007–2010 experience, peer reviewed by The Actuarial Society of Kenya (TASK) and approved by the Insurance Regulatory Authority (IRA), with an updated issue, I believe, now due. Both are real achievements. But both KE 2001-2003 and KE 2007-2010 are assured lives tables, built from life assurance experience to price the risk that policyholders die earlier than expected.

For annuity pricing specifically, the profession has gone a step further. There is a Kenyan annuitant pricing table, graduated using the Brass-Logit method against the UK a(90) as the reference standard, used because local annuitant experience was too thin to stand on its own. That is a real piece of work and the methodology is sound. But the experience underneath the table is still British. What we have is a Kenyan transformation of a foreign basis, not a Kenyan basis. The bottleneck is data, not effort.

That distinction matters. As a pricing actuary writing a guaranteed life annuity in East Africa (EA) today, I am still effectively choosing between bases that are all calibrated to populations other than the one I am underwriting. The local assured lives tables are domestically calibrated but built for the opposite direction of risk. Conservative pricing on a death benefit assumes heavier mortality, which on an annuity quietly understates how long the annuitant will live. South Africa's SAIML98 and SAIFL98 (South African Insured Male/Female Lives 1998) annuitant tables point the right way on risk but come from a foreign, though African, population. The UK annuitant tables, the a(55), the A57, PA(90) and their later cousins, and the a(90) sitting underneath our own graduated table, all point the right way but from an even more distant population. So, most product developers blend. Some at a lower % of a chosen table, some at higher %, some splitting between South African and Kenyan bases, all of it calibrated to portfolio composition rather than to a study of the lives we are actually underwriting.

This is not unreasonable judgment. It is uncalibrated judgment. The adjustments are made without an experience study to tell the actuary whether they are even directionally right, let alone right in size. And there is a layer beyond the one we are missing. The markets that built annuitant tables went a step further and built impaired life tables, the basis that lets an insurer recognise a shorter expected lifespan for someone with a diagnosed condition and pay them a higher income for it. EA has no impaired life table at all. The floor we have rests on a foreign foundation, and the storey above it has not been built.

There is one more disconnect worth naming, though whether it stings depends on who is at the receiving end. In valuation reports to the IRA, where the IFRS 17 best estimate requirement now bites, companies increasingly use more current and sophisticated bases than they apply in commercial pricing. A gap between the pricing basis and the reserving basis is not unusual in a transitioning market, but it is a risk management concern that nobody is systematically tracking.

We are adjusting tables without knowing whether the adjustments are right. I would not call that conservatism. That is navigation by feel in a space that should be mapped.

What other African markets found when they looked is instructive. South Africa's SAIML98 and SAIPL98 work established that its annuitants died materially later than both population tables and assured lives tables predicted, and that Actuarial Society of South Africa's (ASSA) measured improvement of roughly 0.7% per annum for males was likely a floor for the insured population, given socioeconomic selection. Malawi's experience study, completed under Reserve Bank leadership between 2021 and 2024, went further still. Malawian annuitant mortality came in lighter than both the South African annuitant tables and the UK a(90). If that finding repeats in Kenya, and there is no structural reason to assume it would not, then EA insurers are pricing guaranteed annuities on a basis far too heavy for the population they are underwriting. That is not a conservative cushion. It is an understatement of the liability being accepted.

And even if the table were a perfect match for the population today, it would still only be a snapshot. The harder question is what mortality does next. That is where a pricing miss turns into a liability that grows under the contract. Every guaranteed annuity priced on a static table carries an implicit claim that mortality will not change over the life of the contract. For a 25-year payment horizon, that is a claim contradicted by everything we know about how mortality behaves in improving economies. At a 1% annual improvement rate, a conservative figure given Ghana's measured population improvement of 1.24% per annum since 1960, mortality at a given age is roughly 18% lower after 20 years than at the start. The annuity factor is sensitive at the tail. An 18% drop in mortality across ages 70 to 85 makes the annuity factor materially larger than a static basis implies. Across a book in force for 10 years or more, the gap between reserved and actual payments compounds. It is invisible in any single year. It is plain over a generation.

The cohort dimension sharpens this. Research applying Cairns-Blake-Dowd models to Sub-Saharan data finds a pronounced cohort effect, where year of birth matters a great deal for pension age mortality. A person born in 1965, retiring in Nairobi today, has spent an entire working life in an environment of improving healthcare and falling communicable disease. Their mortality at 72 or 78 is not the mortality of any 72 or 78 year old in a historical dataset. The cohort carries its improvement history into retirement. No static table captures that.

IFRS 17 is precise here. Fulfilment cashflows must reflect current best estimate assumptions. Not the pricing assumptions. Not the table in force when the policy was written. To sign off annuity liabilities on a zero-improvement assumption is to certify that mortality will not move across a 25 year horizon, in a country whose life expectancy has risen by 20 years since 1960. That is not a judgment you can defend against the evidence. It has survived only because nothing requires it to be justified and no local study forces the issue. The regulator, if willing, can change that. A simple guidance note, asking appointed actuaries to document and justify their longevity improvement assumption, and to support a zero assumption with formal analysis rather than the absence of a better number, would move the whole market. The cost is low. The governance benefit would compound for generations.

So far everything has been about measurement. How big the promise is, how well we size it, what happens when we get it wrong. The next question shifts ground. Even if every assumption were right, the question would remain whether the assets exist to back the promise. That answer is more public than the mortality answer because it shows up directly in the rate the retiree is offered.

Picture a long-deferred annuity. Say at my age, under 30, I buy a deferred annuity with a 20 year deferment, longer if allowed, followed by a 20 to 25 year payout. The insurer takes my premium and has to find assets that match a liability of 40 to 45 years, earning a spread above the promised rate that covers expenses and leaves a margin, since they are in business, not charity. Kenya's Treasury bond curve now extends to 30 years, which is real progress. Infrastructure bonds, attractive for their tax exemption, have cleared at recent

auctions in tenors of roughly 7 to 15 years. But there is no 40 year bond. So for that deferred annuity the back end of the liability, years 30 through 45, has no matching asset. It gets rolled, reinvested and managed through a duration gap that is structural and, for now, has no market answer.

This shows up directly in declared rates. In 2024 infrastructure bonds yielded 14% to 16% tax free, and short-term Treasury bills peaked at 16% to 17% in mid-2024 before falling sharply as the Central Bank of Kenya (CBK) eased, with the 91 day bill below 9% by early 2025. The 10 year government bond yield, near 19.4% in April 2024, had eased to roughly 11% to 13% by late 2025. Against that backdrop, industry reports through 2024 placed declared annuity rates in roughly the 11% to 13% range at peak. The maths is not subtle. Government paper was paying 14% to 16% while annuity rates ran two to four points below it. The insurers were not pocketing excess margin. They were pricing for the reinvestment risk they carry across four decades, when their longest asset matures 15 years before the liability ends. The retiree pays for the duration gap in the form of a suppressed rate. That one is not on the actuaries. It is a capital market infrastructure failure, and yet someone has to wear it. I am afraid that someone is the retiree.

The UK solved a version of this by issuing gilts at 2, 5, 10, 20, 30 and 50 year maturities. The 50 year gilt exists precisely because pension funds and annuity writers asked for it. Solvency UK, live from December 2024, goes further with a Matching Adjustment that rewards alignment. Hold long dated assets matched to your annuity cashflows and you may take a credit spread premium in your discount rate, which lowers your capital requirement. A matched book carries less reinvestment and liquidity risk than an unmatched one, and the rules reflect it. Kenya has neither the long bond nor the Matching Adjustment, and there is no domestic alternative solving the problem.

The asset gap has another face. Even if the duration matched, the income still has to keep up with inflation, and that is a separate asset problem we have not solved. A level annuity in a 6% inflation environment loses half its real value in 12 years. Our escalation menu, for lack of a better word, runs level, fixed 3%, 5%, occasionally 7%, and none of it responds to actual inflation. When Kenyan Consumer price index (CPI) sat above 7% in 2022 and 2023, a 3% escalating annuity was losing four to five points in real terms, in a product sold as stable retirement income. As a future retiree, that does not sound good to me. Real protection needs inflation linked assets Kenya does not issue, the same duration gap wearing a different coat. South Africa's answer is instructive. Instead of full CPI linking, which it also struggles to hedge, the market starts many annuities at 0% escalation and actively smooths the in-force book, concentrating real income early in retirement when needs are highest. EA has neither the products nor the advice framework at the point of retirement that would let a buyer weigh these options against their own life expectancy.

And those are just the gaps on the supply side. There is another problem one layer deeper, which is that even a properly priced, properly hedged, properly escalating annuity may still not be the product the EA retiree actually wants. The low uptake here is sometimes blamed on financial literacy. I think that's a misplaced blame. It is a rational response to a product that does not solve the problem the EA retiree actually has. The retiring formal sector worker is usually the main earner for a web of dependants. Holding on to capital is not a failure to grasp compound interest. It is the knowledge that the premium is locked away the moment it is paid, that nothing returns to the family on death and that no emergency can be met from a monthly cheque. The fixes are not exotic. Period certain guarantees, joint life cover and tiered designs that annuitize only part of the pot. They lift uptake without adding much longevity cost, and they need product development and regulatory clarity, both of which we can muster.

So, now what? The mortality basis is a call each appointed actuary makes. The interim is simple in theory if a stretch in practice. Move to SAIML98 and SAIFL98 as a base, layer on the Malawi scalings now public, and apply an explicit improvement scale of 0.7% to 0.9% per annum, roughly 60% to 70% of Ghana's population rate to reflect faster improvement among the insured. IFRS 17 sensitivity disclosures should already show the reserve impact from zero up to 1.5%. None of these needs new data or new law. What stops it is competitive dynamics. If one company moves and the rest do not, the mover faces higher prices or thinner margins, and nobody wants to be the sacrificial lamb. The regulator can break that deadlock with a minimum standard for annuity pricing assumptions, including a requirement that a zero-improvement assumption be formally justified rather than quietly adopted.

The bigger prize is an EA annuitant experience study built from local data. TASK and the IRA have been progressing this work through the last year, and the binding constraint, as I understand it, has been data scarcity. That is exactly why pooled industry data is the unlock. The structure is straightforward. TASK, the IRA and the major writers pool anonymised in-force data, entry ages, policy anniversaries and death claims, for one joint investigation. Rwanda did it with Barnett Waddingham and Callund Consulting in under three years, with an explicit brief to build local capacity. Malawi did it in house. The data already sits in six to eight companies. Pooled, the study becomes tractable, and the only real obstacle is collective action, which is exactly what TASK exists to solve. Alongside it, medical underwriting for enhanced annuities is the product with the clearest commercial case, the impaired life pricing I mentioned earlier finally put to work. Reinsurance Group of America (RGA) and Munich Re already run African enhanced annuity frameworks, and a short health questionnaire is enough to segment the pool and pass real value to the retirees who need it most.

The duration gap is the one the industry cannot fix alone. Extending the curve to 30, then 40 and 50 years needs a collective case, made by TASK, the Retirements Benefits Authority (RBA), the AKI and the major fund managers, to the National Treasury, with the macro argument that a deep domestic annuity market keeps long term capital at home and helps fund infrastructure, which serves the fiscus as much as the industry. Inflation linked issuance follows from the same constituency making the same case. Both are ordinary instruments in markets facing the volatility we face. They need political will and a willing issuer. The industry is the natural demand side. The conversation has been raised in pockets, including at recent TASK convention sessions, but it has not yet converged into a coherent ask to Treasury.

None of the above absolves the individual appointed actuary. Even ahead of any industry-wide study, the professional duty to run experience investigations on one's own book, and to adjust the table accordingly for both pricing and valuation, is the operational bridge between an inadequate industry standard and a solvent company. The standard is the floor. The company actuary's experience analysis is what determines whether the building above it stands.

What makes this moment different from the same speech 10 years ago is IFRS 17. Its demand for best estimate liabilities, explicit risk adjustments and documented assumptions makes quietly carrying a static basis much harder. The appointed actuary's signature now means more than it used to, and that is a fair pressure to feel. It also arrives at exactly the wrong moment to ignore. Kenya's retirement industry is growing fast enough that, within a decade, today's actuarial infrastructure will not cope at scale. Pension assets already approach 15% of GDP and keep climbing, and the annuity book will follow a wave of formal sector retirees who, for the first time, hold meaningful capital. The tables need updating before the book is large enough for the error to be systemic. The improvement scale needs to leave zero before a generation of static pricing matures into a shortfall. The duration gap needs attention before the market is big enough for a rate shock to become a solvency event. Each one is manageable now and less so every year.

On the other side of every one of these decisions is a retiree who fed the system for 35 years and gets one shot at turning that into an income they can live on. What our pricing infrastructure delivers, or fails to, decides whether the promise of retirement security in East Africa is real or just performative.